

Problem

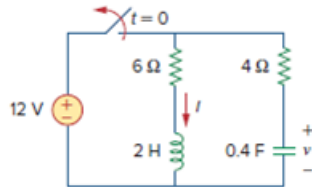
For the circuit in Fig. 8.62, find:

(a) $i(0^+)$ and $v(0^+)$,

(b) $di(0^+)/dt$ and $dv(0^+)/dt$,

(c) $i(\infty)$ and $v(\infty)$.

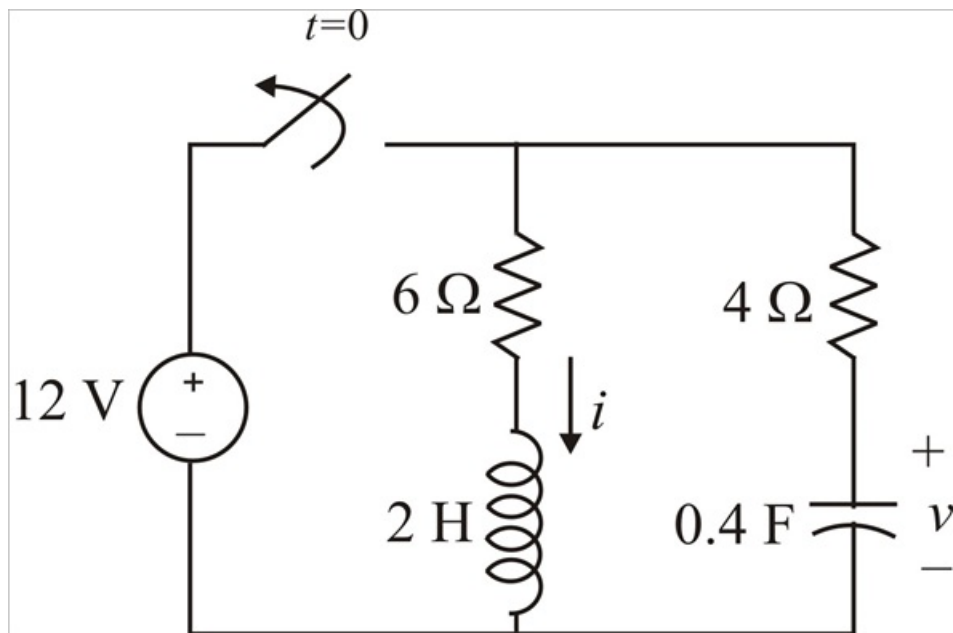
Figure 8.62:



Step-by-step solution

Step 1 of 8

Given circuit diagram

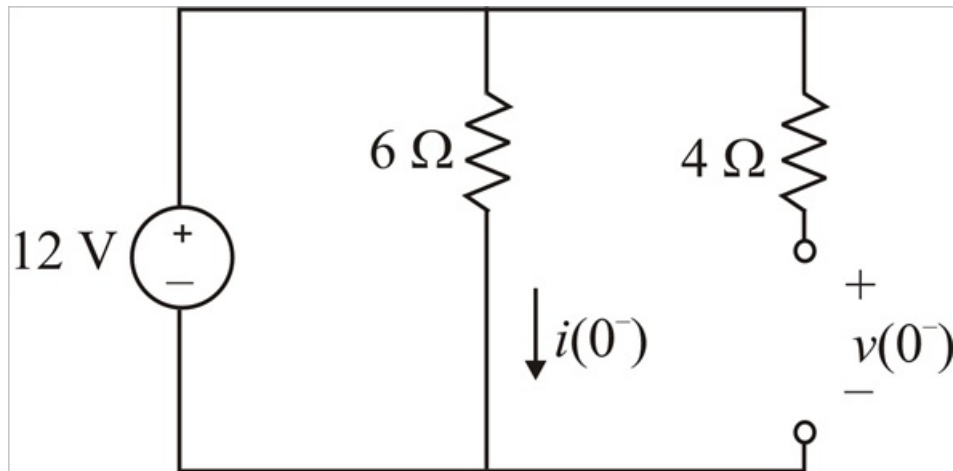


Step 2 of 8

(a)

For $t < 0$ switch is in open condition and circuit is acts under steady state condition

At $t = 0^-$ inductor acts like a short circuit and capacitor acts like a open circuit hence given circuit is as below



Step 3 of 8

From the above circuit we can write

$$i(0^-) = \frac{12}{6}$$

$$i(0^-) = 2\text{A}$$

Inductor current can't change instantaneously

$$i(0^-) = i(0^+) = 2\text{A}$$

$$\boxed{i(0^+) = 2\text{A}}$$

From the above circuit we can write

$$v(0^-) = [i(0^-)](6)$$

$$v(0^-) = (2)(6)$$

$$v(0^-) = 12\text{V}$$

Capacitor voltage can't change instantaneously

$$v(0^-) = v(0^+) = 12\text{V}$$

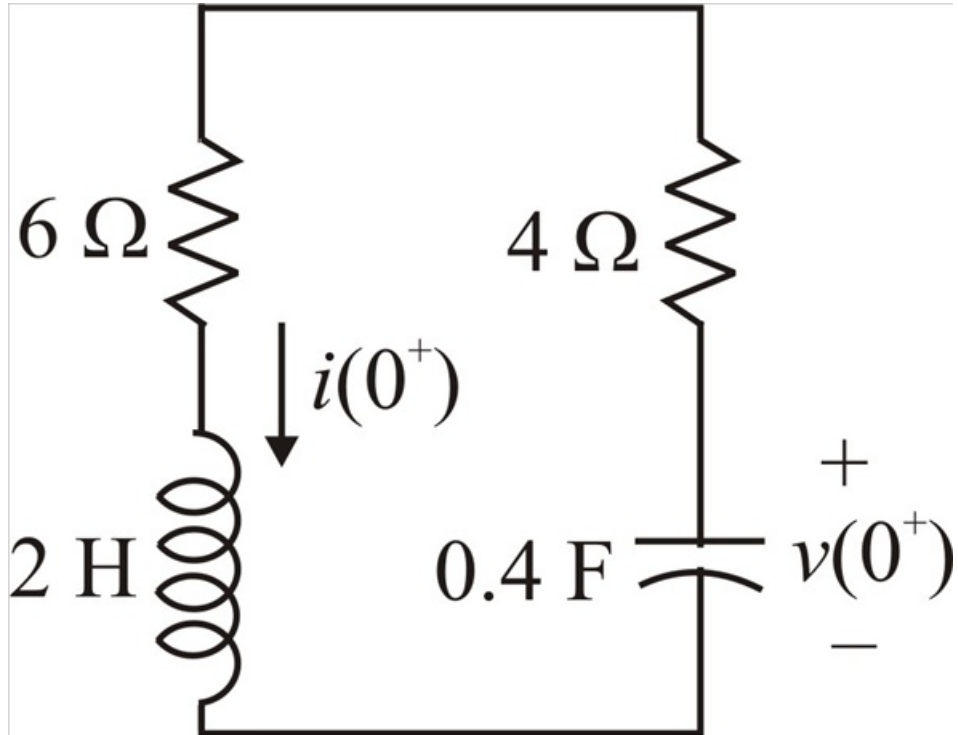
$$\boxed{v(0^+) = 12\text{V}}$$

Step 4 of 8

(b)

Inductor current and capacitor voltage can't change instantaneously.

Hence at $t = 0^+$ given circuit is as below



Step 5 of 8

By applying KVL to the above circuit we can write

$$-v(0^+) + [i(0^+)](4) + [i(0^+)](6) + v_L(0^+) = 0$$

$$-12 + (2)(4) + (2)(6) + v_L(0^+) =$$

$$-12 + 8 + 12 + v_L(0^+) = 0$$

$$v_L(0^+) = -8V$$

Step 6 of 8

Equation for inductor voltage

$$v_L(t) = L \frac{di_L(t)}{dt}$$

At $t = 0^+$

$$v_L(0^+) = L \frac{di_L(0^+)}{dt}$$

$$\frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L}$$

$$\frac{di_L(0^+)}{dt} = \frac{-8}{2}$$

$$\boxed{\frac{di_L(0^+)}{dt} = -4A/s}$$

Step 7 of 8

Equation for capacitor current

$$i_c(t) = C \frac{dv_c(t)}{dt}$$

At $t = 0^+$

$$i_c(0^+) = C \frac{dv_c(0^+)}{dt}$$

$$\frac{dv_c(0^+)}{dt} = \frac{i_c(0^+)}{C}$$

From the above circuit

$$i_c(0^+) = -i(0) = -2\text{A}$$

$$\frac{dv_c(0^+)}{dt} = \frac{-2}{0.4}$$

$$\frac{dv_c(0^+)}{dt} = -5\text{V/s}$$

We know that $v_c(0^+) = v(0^+)$

$$\boxed{\frac{dv(0^+)}{dt} = -5\text{V/s}}$$

Step 8 of 8

(c)

For $t > 0$ given circuit is a source free network

Hence at $t = \infty$ voltage and current are

$$\boxed{i(\infty) = 0\text{A}} \text{ and}$$

$$\boxed{v(\infty) = 0\text{V}}$$